

DEVOPS PROGRAM 3:

Code a Dockerized python Flask or [Node.js](#) Application.

PROJECT STRUCTURE:

- program3/
- Dockerfile(without any extension)
- requirement.txt
- app.py

Step 1: Create Project Directory

`mkdir prog3`

Creates a new directory named program3

Step 2: Navigate to the Project Directory

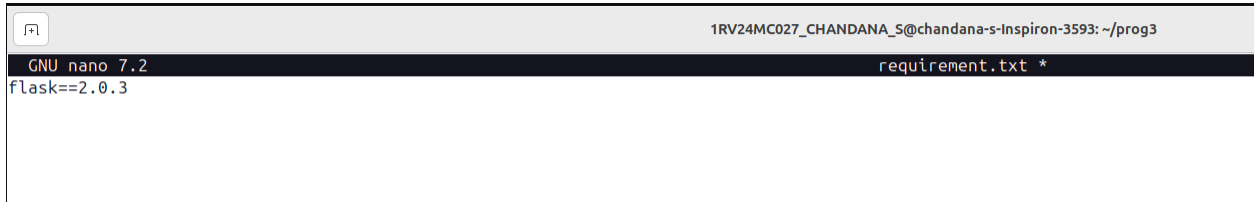
`cd prog3`

Changes the working directory to program1, where all files will be created.

```
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ cd prog3
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/prog3$ nano requirement.txt
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/prog3$ nano app.py
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/prog3$ nano Dockerfile
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/prog3$ docker build -t prog3 .
[+] Building 2.7s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 470B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 1.1s
=> [internal] load .dockerignore                                   0.2s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338 0.3s
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338 0.3s
=> [internal] load build context                                   0.2s
=> => transferring context: 753B                                       0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> CACHED [3/5] COPY requirement.txt .                             0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirement.txt 0.0s
=> CACHED [5/5] COPY . .                                           0.0s
=> exporting to image                                              0.1s
=> => exporting layers                                              0.0s
=> => writing image sha256:bc317b37b93d803acdce61fdb2a48bff2f6652ae55a1d 0.0s
=> => naming to docker.io/library/prog3                             0.0s
```

Step 3: Create and Edit requirement.txt file

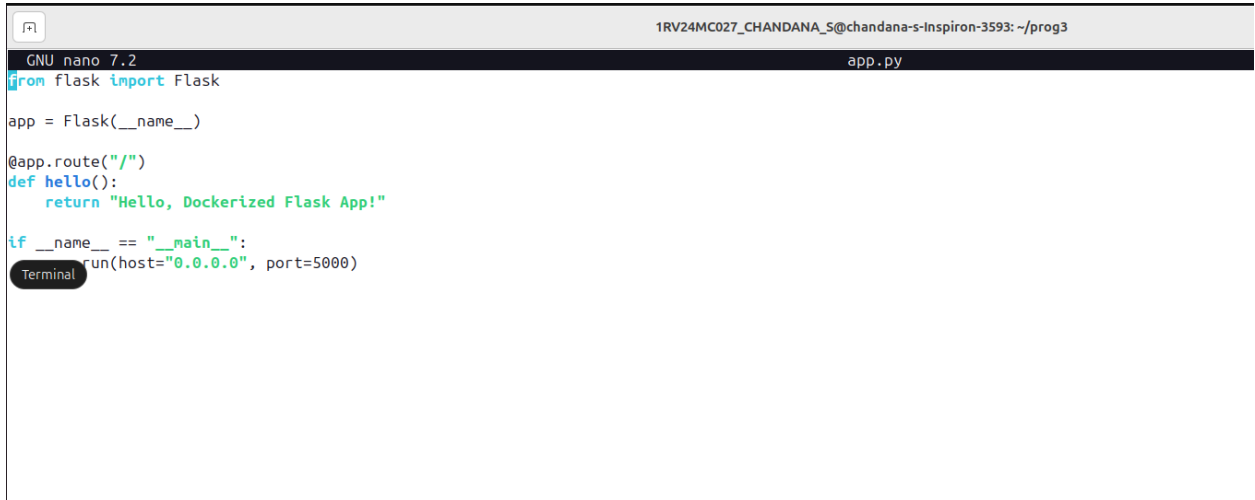
nano requirement.txt

A screenshot of a terminal window showing the nano text editor editing a file named requirement.txt. The top status bar shows the user is 1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593 in the directory ~/prog3. The editor header shows 'GNU nano 7.2' and 'requirement.txt *'. The file content contains two lines: 'flask==2.0.3' followed by a blank line.

```
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593: ~/prog3
GNU nano 7.2 requirement.txt *
flask==2.0.3
```

Step 4: Create and Edit app.py file

nano [app.py](#)

A screenshot of a terminal window showing the nano text editor editing a file named app.py. The top status bar shows the user is 1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593 in the directory ~/prog3. The editor header shows 'GNU nano 7.2' and 'app.py'. The file content contains Python code for a Flask application. A 'Terminal' button is visible in the bottom left corner.

```
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593: ~/prog3
GNU nano 7.2 app.py
from flask import Flask

app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello, Dockerized Flask App!"

if __name__ == "__main__":
    run(host="0.0.0.0", port=5000)
```

Step 5: Create and Edit Dockerfile

nano Dockerfile

```
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593: ~/prog3
GNU nano 7.2 Dockerfile
# This is a Dockerfile

# Use a Python image
FROM python:3.9-slim

# Set the working directory inside the container
WORKDIR /app

# Copy the requirements file and install dependencies
COPY requirement.txt .
RUN pip install --no-cache-dir -r requirement.txt

# Copy the application code
COPY . .

# Expose the port the Flask app will run on
EXPOSE 5000

# Define the command to run the Flask application
CMD ["python", "app.py"]
```

Step 6: Build docker image

```
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/prog3$ docker build -t prog3 .
[+] Building 2.7s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.1s
=> => transferring dockerfile: 470B                               0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 1.1s
=> [internal] load .dockerignore                                  0.2s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338 0.3s
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338 0.3s
=> [internal] load build context                                  0.2s
=> => transferring context: 753B                                       0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> CACHED [3/5] COPY requirement.txt .                             0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirement.txt 0.0s
=> CACHED [5/5] COPY . .                                           0.0s
=> exporting to image                                              0.1s
=> => exporting layers                                              0.0s
=> => writing image sha256:bc317b37b93d803acdce61fdb2a48bff2f6652ae55a1d 0.0s
=> => naming to docker.io/library/prog3                             0.0s
```

Step 7: Run the docker container with specifying port number

```
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/prog3$ docker run -p 5000:5000 prog3
* Serving Flask app 'app' (lazy loading)
* Environment: production
  WARNING: This is a development server. Do not use it in a production deployment.
  Use a production WSGI server instead.
* Debug mode: off
* Running on all addresses.
  WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://172.18.0.2:5000/ (Press CTRL+C to quit)
```

Step 8: open browser and run



Step 9: Stop and remove docker container and docker images

```
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ docker container ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
d18cd9991765   prog3          "python app.py"         6 minutes ago   Exited (0)    3 minutes ago   admiring_austin
190a29c4312e   bc317b37b93d   "python app.py"         9 minutes ago   Exited (1)    9 minutes ago   practical_yalow
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ docker container rm d18 190
d18
190
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
prog3         latest   bcde523ecba1   7 minutes ago   132MB
<none>        <none>   bc317b37b93d   11 minutes ago   132MB
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ docker rmi bcd bc3
Untagged: prog3:latest
Deleted: sha256:bcde523ecba1ca2d9e028a92f8343a93d16f818e1622973ddb44d6db1148c6c
Deleted: sha256:bc317b37b93d803acdce61fdb2a48bff2f6652ae55a1d520604d044c341a67cd
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$
```